

MAJOR PROJECT

PROJECT TITLE : 1. Network Scanning and
Port Scanning.
2. Password Strength Checker
Of windows.

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SUBMITTED BY : SkilForge Company

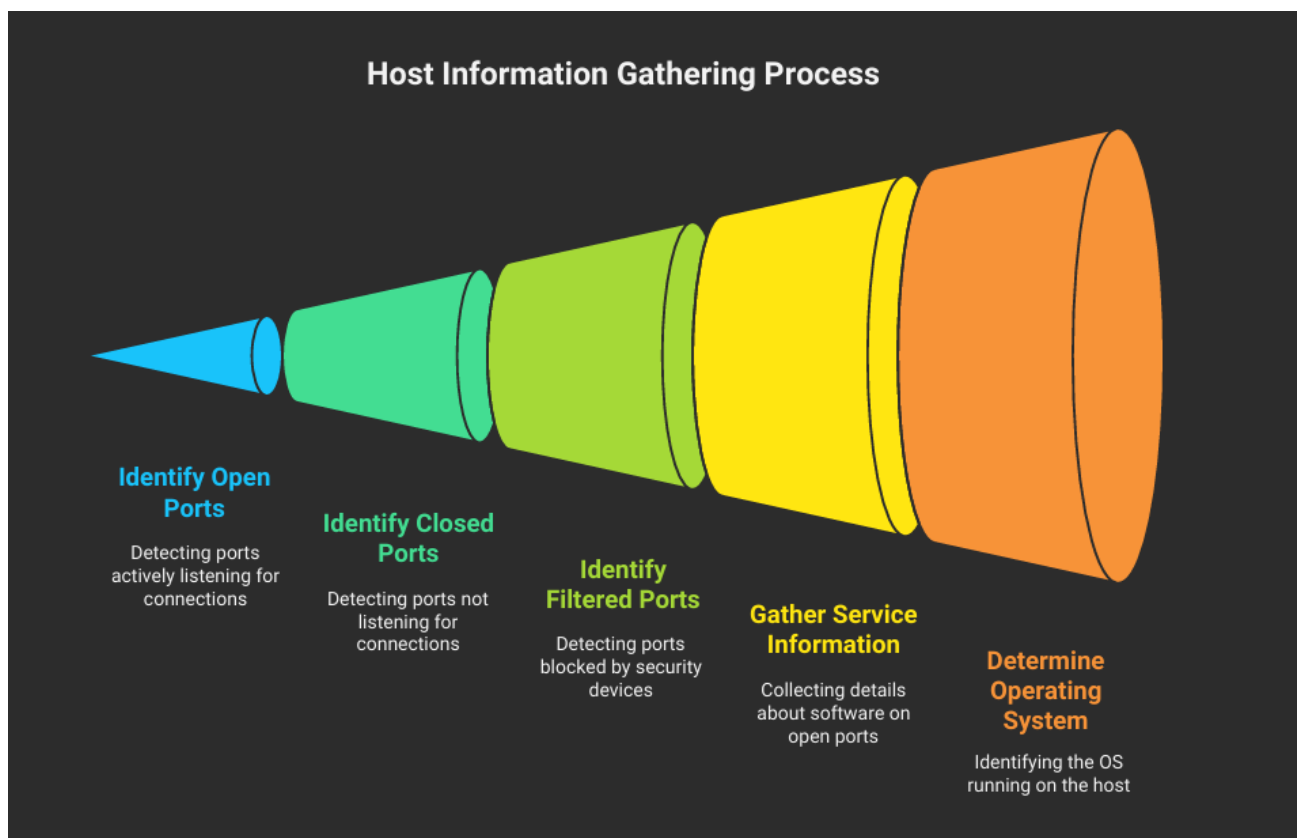
DATE OF : 04/10/2025

MISSION

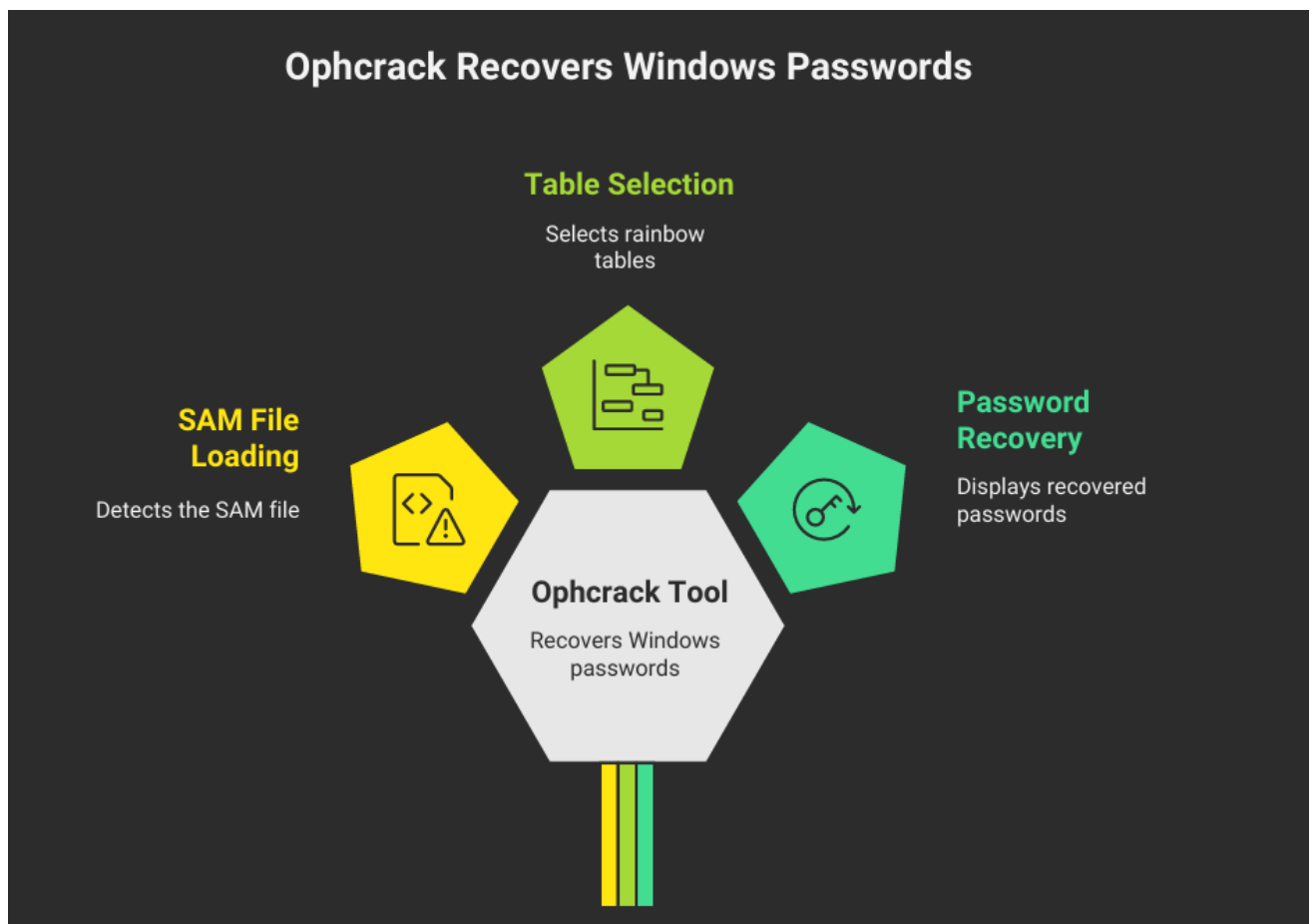
Introduction :

This Project is for practice of protecting systems, networks, and programs from digital attacks. Vulnerabilities such as exposed ports or weak passwords allow attackers to gain unauthorized access, steal data, or disrupt services. This project focuses on two common areas of exploitation:

(1) Network and Port Scanning to discover exposed services.



(2) Offline Password recovery to demonstrate the risk of weak local account passwords. All activities were conducted in an isolated virtual lab to ensure ethical and safe experimentation.



Objectives :

The main objectives of this project are:

1. To perform network and port scanning on target virtual machines using Nmap.
2. To identify open ports, services, and service versions that

3. To recover or test Windows 7 local account passwords using Ophcrack to show the impact of weak passwords.
4. To explain the Windows SAM file and its role in storing password hashes.

Tools Required :

- NMAP :

Nmap (Network Mapper) is an open-source tool for network discovery and security auditing. It can discovery hosts, open ports, services, and versions, and run NSE (Nmap scripting Engine) scripts to gather additional information.

- OPHCRACK :

Ophcrack is an open-source password cracking that uses rainbow tables to reverse LM/NTLM password hashes into plaintext. It is effective against weak and common passwords ; Complex passwords may resist recovery.

TASK 1 : Network & port scanning (Nmap)

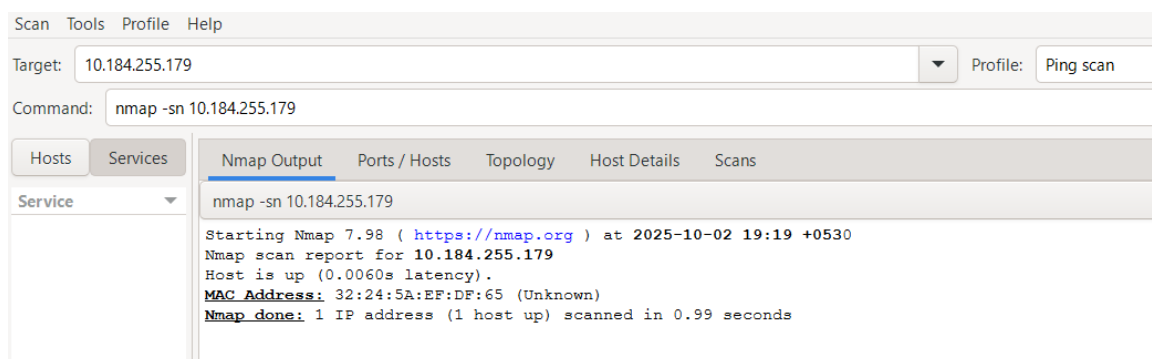
To identify active hosts, discover open ports, and detect the services and versions running on the target machines. This helps to highlight exposed services that can be prioritized for hardening.

LINUX

Tools & Commands :

1. Ping Scan :

- Definition : A ping scan is used to check whether a host (computer, server, or device) on a network is alive (active) or dead (inactive).
- Purpose : To discover active devices in a network.
- Command : `nmap -sn 10.184.255.179`



➤ The Ping scan showing one host is alive (active).

2.Intense scan :

- **Definition** : An Intense Scan is a comprehensive Nmap scan that not only checks if hosts are alive but also performs detailed port scanning, service detection, OS detection, and traceroute. It gives a full picture of the target system.
- **Purpose** : 1. To detect open ports and the services running on them.
2. To identify the operating system and network characteristics.
- **Command** : `nmap -T4 -A -v 10.184.255.179`.

Target:	10.184.255.179	▼	Profile:	Intense scan
Command:	nmap -T4 -A -v 10.184.255.179			
Hosts	Services	Nmap Output		
Service	domain			
		Hostname	Port	Protocol
		10.184.255.179	53	tcp
				open
				dnsmasq 2.51

```
PORT      STATE SERVICE VERSION
53/tcp    open  domain dnsmasq 2.51
```

- The Intense Scan revealed only one open port – TCP 53, running dnsmasq version 2.51 and Host name is 10.184.255.179.

3.Intense scan plus UDP :

- Definition : A Scan that checks both TCP and UDP ports to detect active services, their versions, and the operating system.
- Purpose : To gather detailed information about all network services, including UDP-based ones, for security assessment.
- Command : `nmap -sS -sU -T4 -A -v 10.184.255.179`

▼ 10.184.255.179

▼ **Host Status**

State: up

Open ports: 1

Filtered ports: 0

Closed ports: 999

Scanned ports: 1000

Up time: 1627804

Last boot: Fri Sep 12 22:18:38 2025

▼ **Addresses**

IPv4: 10.184.255.179

IPv6: Not available

MAC: B2:E5:B4:1A:AC:43

▼ **Operating System**

Name: Android 9 (Linux 4.9)

Accuracy:

▼ **Ports used**

Port- 53 -
Protocol- tcp -
State: open

Port- 1 - tcp
Protocol- -
State: closed

Port- 44441
Protocol- - udp
State: - closed

▼ **OS Classes**

Type	Vendor	OS Family	OS Generation	Accuracy
phone	Linux	Linux	4.X	

▼ **TCP Sequence**

Difficulty: Good luck!

Index: 259

Values:

▼ **IP ID Sequence**

Class: All zeros

Values:

▼ **TCP TS Sequence**

Class: 1000HZ

Values:

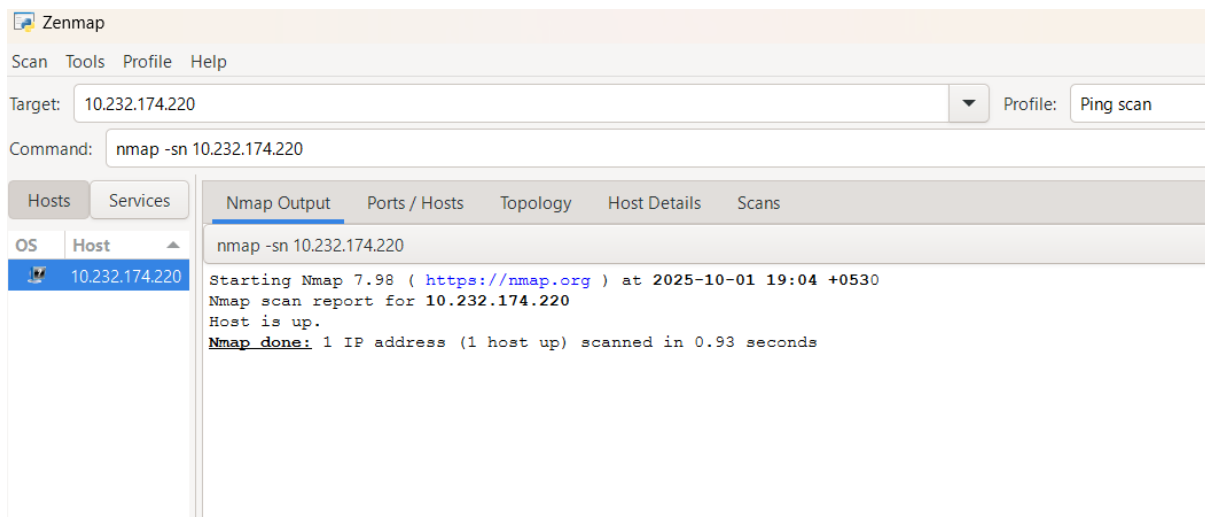
- The Itense scan plus UDP, gives all TCP & UDP ports
From the active host which is 10.184.255.179.

WINDOWS

Tools & Commands :

1. Ping Scan :

- Definition : A ping scan is used to check whether a host (computer, server, or device) on a network is alive (active) or dead (inactive).
- Purpose : To discover active devices in a network.
- Command : `nmap -sn 10.232.174.220`



- The Ping scan showing one host is alive (active).

2.Intense scan :

- Definition : An Intense Scan is a comprehensive Nmap scan that not only checks if hosts are alive but also performs detailed port scanning, service detection, OS detection, and traceroute. It gives a full picture of the target system.
- Purpose : 1. To detect open ports and the services running on them.
2. To identify the operating system and network characteristics.
- Command : `nmap -T4 -A -v 10.232.174.220`

```
Nmap Output  Ports / Hosts  Topology  Host Details  Scans
nmap -T4 -A -v 10.232.174.220

Nmap scan report for 10.232.174.220
Host is up (0.00067s latency).
Not shown: 978 closed tcp ports (reset)
PORT      STATE  SERVICE          VERSION
25/tcp    filtered smtp
110/tcp   filtered pop3
119/tcp   filtered nntp
125/tcp   filtered locus-map
135/tcp   open    msrpc             Microsoft Windows RPC
139/tcp   open    netbios-ssn       Microsoft Windows netbios-ssn
143/tcp   filtered imap
445/tcp   open    microsoft-ds?
465/tcp   filtered smtps
548/tcp   filtered afp
563/tcp   filtered snews
587/tcp   filtered submission
800/tcp   filtered mdbas_daemon
902/tcp   open    ssl/vmware-auth   VMware Authentication Daemon 1.10 (Uses VNC, SOAP)
903/tcp   filtered iss-console-mgr
912/tcp   open    vmware-auth       VMware Authentication Daemon 1.0 (Uses VNC, SOAP)
993/tcp   filtered imaps
995/tcp   filtered pop3s
1433/tcp  filtered ms-sql-s
3306/tcp  open    mysql             MySQL 8.0.32

8383/tcp  open    ssl/http          nginx
8443/tcp  open    ssl/https-alt
```

Target: 10.232.174.220

Profile: Intense scan

Command: nmap -T4 -A -v 10.232.174.220

HostsServices

OS Host

10.232.174.220

Nmap Output		Ports / Hosts		Topology	Host Details	Scans
	Port	Protocol	State	Service	Version	
🚩	25	tcp	filtered	smtp		
🚩	110	tcp	filtered	pop3		
🚩	119	tcp	filtered	nnntp		
🚩	125	tcp	filtered	locus-map		
🟢	135	tcp	open	msrpc	Microsoft Windows RPC	
🟢	139	tcp	open	netbios-ssn	Microsoft Windows netbios-ssn	
🚩	143	tcp	filtered	imap		
🟢	445	tcp	open	microsoft-ds		
🚩	465	tcp	filtered	smtps		
🚩	548	tcp	filtered	afp		
🚩	563	tcp	filtered	snews		
🚩	587	tcp	filtered	submission		
🚩	800	tcp	filtered	mdb_s_daemon		
🟢	902	tcp	open	vmware-auth	VMware Authentication Daemon 1.10 (Uses VNC, SOAP)	
🚩	903	tcp	filtered	iss-console-mgr		
🟢	912	tcp	open	vmware-auth	VMware Authentication Daemon 1.0 (Uses VNC, SOAP)	
🚩	993	tcp	filtered	imaps		
🚩	995	tcp	filtered	pop3s		
🚩	1433	tcp	filtered	ms-sql-s		
🟢	3306	tcp	open	mysql	MySQL 8.0.32	
🟢	8383	tcp	open	http	nginx	
🟢	8443	tcp	open	https-alt		

➤ The Intense scan plus UDP, gives all TCP & UDP ports

From the active host which is 10.232.174.220.

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- Definition : A Scan that checks both TCP and UDP ports to detect active services, their versions, and the operating system.
- Purpose : To gather detailed information about all network services, including UDP-based ones, for security assessment.
- Command : `nmap -sS -sU -T4 -A -v 10.184.255.179`

▼ 10.232.174.220

▼ Host Status

State: up

Open ports: 8



Filtered ports: 14

Closed ports: 978

Scanned ports: 1000

Up time: 437183



Last boot: Fri Sep 26 17:39:11 2025

▼ Addresses

IPv4: 10.232.174.220

IPv6: Not available

MAC: Not available

▼ Operating System

Name: Microsoft Windows 10
1607 - 11 23H2

Accuracy:

▼ Ports used

Port- 135 -
Protocol- tcp -
State: open

Port- 1 - tcp
Protocol- -
State: closed

Port- 37615 - udp
Protocol- -
State: closed

▼ OS Classes

Type	Vendor	OS Family	OS Generation	Accuracy
general purpose	Microsoft	Windows	11	

▼ TCP Sequence

Difficulty: Good luck!

Index: 262

Values:

▼ IP ID Sequence

Class: Incremental

Values:

▼ TCP TS Sequence

Class: 1000HZ

Values:

- The Itense scan plus UDP, gives all TCP & UDP ports
From the active host which is 10.184.255.179.

Output :

- Nmap host discovery identified live machines on the local lab network.
- On the Windows 11 , open ports such as 563 (TCP) and 912 (TCP) were detected.
- On the Linux , ports such as 53 TCP were discovered.
- Service/version detection showed outdated versions of some services (e.g., SMB v1, Apache 2.x).

Conclusion :

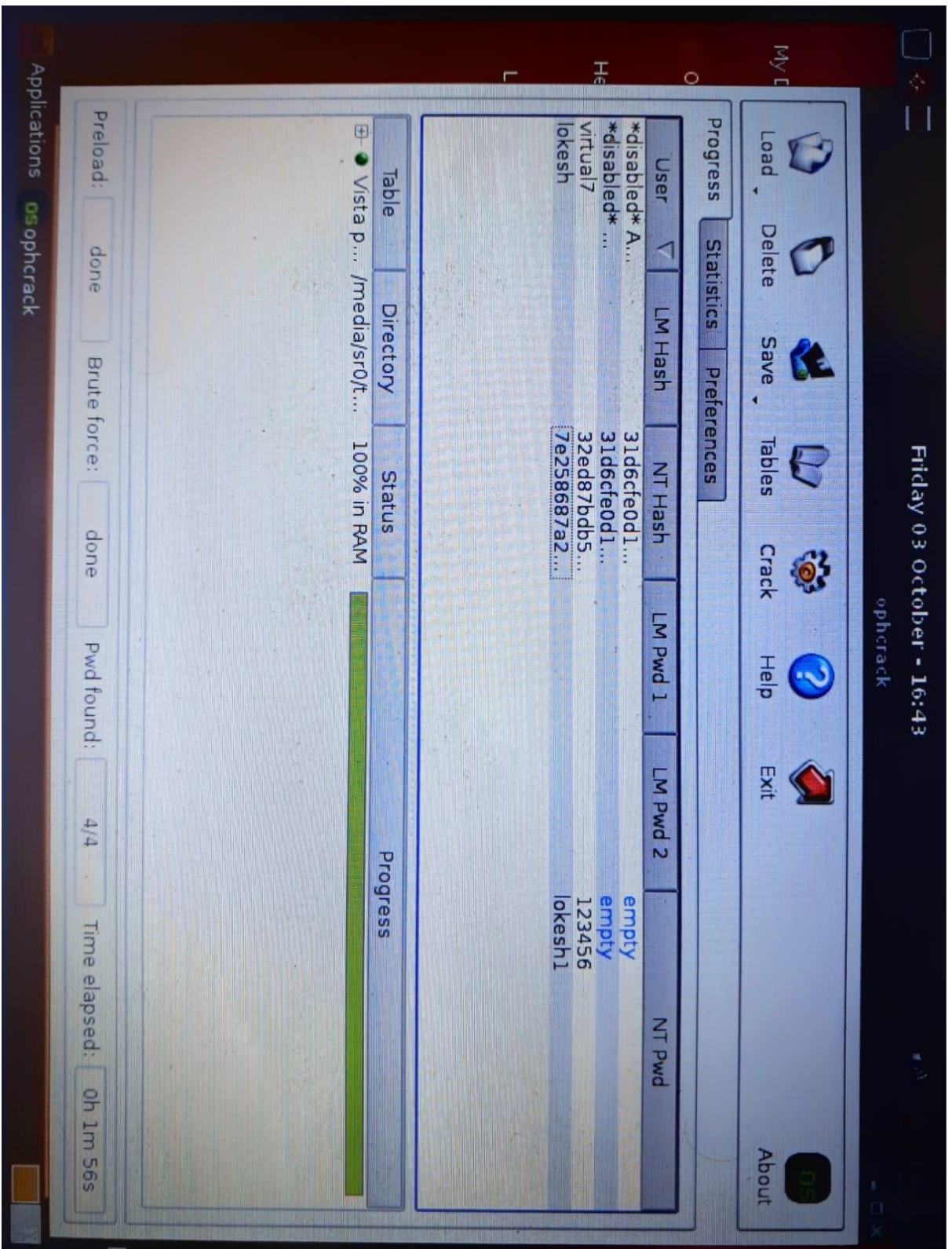
Task 1 successfully demonstrated how Nmap can be used to map active hosts, detect open ports, and identify running services. The exercise proved that exposed services increase risk if not managed properly. Hardening measures such as closing unused ports, patching software, and restricting access through firewalls are necessary to reduce vulnerabilities.

TASK 2 : PASSWORD STRENGTH TESTING (OPHCRACK)

Introduction : This task demonstrates offline password recovery on a windows 7 virtual machine using Ophcrack, a free rainbow-table-based password recovery tool. The purpose is to show how weak local passwords can be recovered from the SAM (Security Accounts Manager) data and to recommend steps to prevent such compromises.

1.SAM File : Windows stores local account hashes in C:\windows\system32\config\SAM. The system hive is also required to decrypt some data. The SAM is locked when windows runs, so offline extraction or a LiveCD approach is used.

2.Ophcrack & rainbow tables : Ophcrack uses precomputed rainbow tables (e.g., “ Vista free tables “) to reverse NTLM/LM hashes back into plaintext quickly when passwords are common or simple. Ophcrack is free and widely used for educational and recovery purpose.



Accounts	Passwords
1. Virtual 7	123456
2. lokesh	lokes1

Explanation :

In a local lab environment running Windows 7, I tested two local administrator accounts — virtual7 and lokesh — using the free tool Ophcrack with the Vista free rainbow tables. Ophcrack quickly recovered both account passwords (123456 for virtual7 and lokesh1 for *lokesh`), showing they were weak and commonly used. Because rainbow tables contain precomputed hashes for common passwords, the tool matched and revealed these passwords almost instantly. This demonstrates that weak local passwords can be recovered offline in minutes if an attacker gains access to the SAM/SYSTEM data or boots from a live media.

Recommendations (to avoid offline password cracking) :

- Use strong password or passphrases (minimum 12 characters, mix of uppercase, lowercase, numbers, special characters).
- Apply account lockout policies and multi-factor authentication for administrator accounts.

Output :

- Two local administrator accounts were tested: virtual7 and lokesh.
- Using Ophcrack with Vista free rainbow tables, both accounts' passwords were successfully recovered :
- virtual7 → 123456
- lokesh → lokesh1

Conclusion : Task 2 successfully showed the risk of weak local passwords in Windows 7. By using Ophcrack, common passwords were revealed quickly, proving that offline recovery is a major threat in case of disk compromise. Strong password policies, disabling LM hashes, enabling full-disk encryption, and applying account lockout/MFA (Multi factor authentication) are essential to protect against offline attacks.